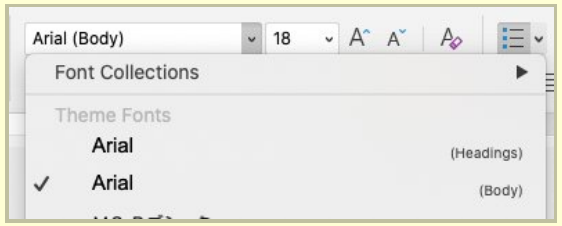
>>>>> Before you begin <<<<<

Capitalization guidelines

- Use **title case** for presentation titles
  - All words are capitalized except for prepositions, conjunctions, and articles
- Use **sentence case** for slide titles, section titles, and bullets
  - Just the first word in a text string is capitalized
- Always **capitalize proper nouns**, including Esri product names as noted in the Product Naming Guide

Font guidelines

- Always pick **Arial** [Headings] or [Body] from the **top** of the font list under Theme Fonts



>>>>> Before you begin <<<<<



Less is more

- Avoid paragraphs and long sentences
- Use the 5x5 rule — max of 5 lines, 5 words each
- One idea per slide



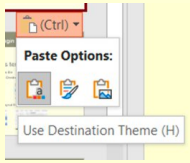
Presentation structure

- Content in longer presentations should be grouped using the following structure:
  - Sections > Topics > Subtopics
  - Note: A section may include multiple topics

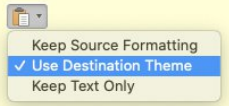
>>>>> Before you begin <<<<<

How to apply this template

1. Copy existing slides into this file
2. When pasting, choose Use Destination Theme



Windows

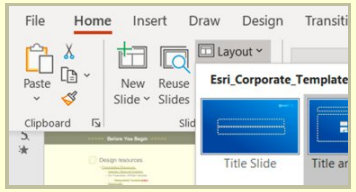


Mac

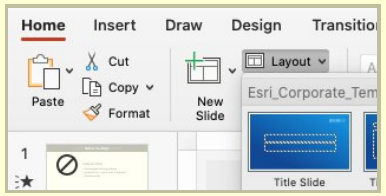


Web

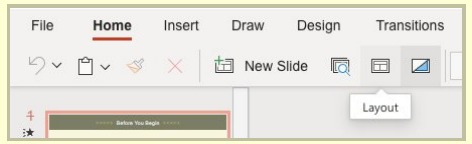
3. For consistent title placement, you may need to select a layout from the Layout menu



Windows



Mac



Web

>>>>> Before you begin <<<<<

Design resources

- Icons
  - From the Home ribbon above, click Esri Icons

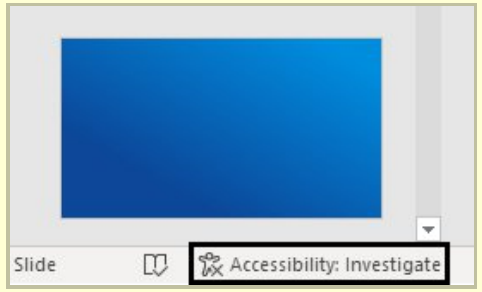


- Presentation Resources [link]
  - Diagrams, Tables and Timelines [link]
  - Esri Design Template [link]
  - Globes [link]
  - Backgrounds [link]
- Strategic Presentations [link]
- Stock photos
  - Adobe Stock images are available. See the details
  - **Do not use** stock images from Microsoft.

>>>>> Before you begin <<<<<

Accessibility [link]

- Check accessibility while designing slides with the Review tab or in the lower left-hand corner
- Add **alternative text** and mark graphical images as decorative
- Review slide **reading order**
- Create **unique slide titles**





Performant maps, Open data

Optimizing large municipal queries with MapLibre

George Owen

Agenda

- Municipal open data
- Browser-based data rendering
- Optimizing GeoJSON queries

Municipal open data

Finding open data

Focus of today's talk

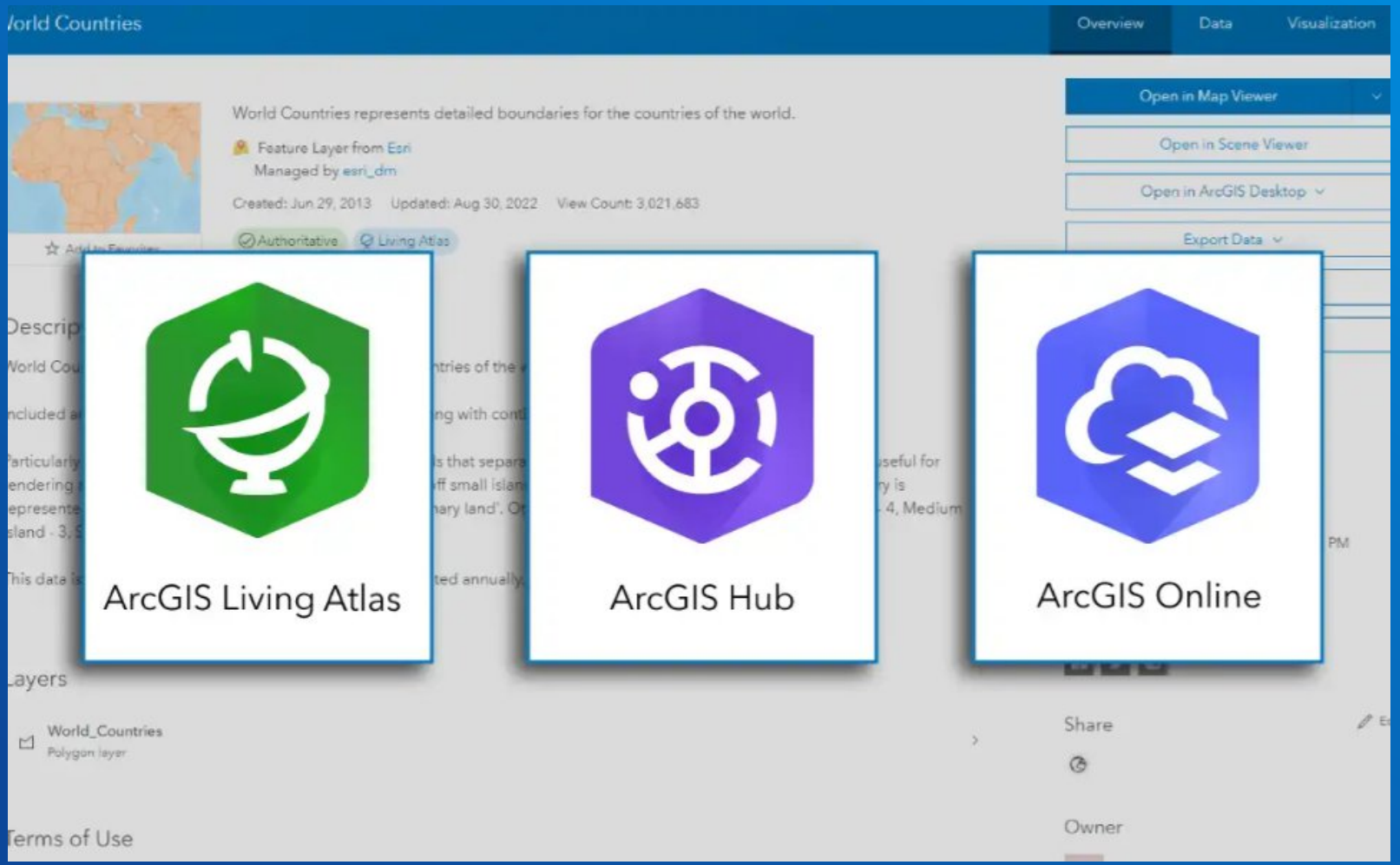
- **ArcGIS Hub:** Municipal data, local gov't
- Today's data: **City of Sacramento**
- Data sources used in today's talk:
  - City of Sacramento Open Data Hub
  - Sacramento County Open Data Hub
  - ArcGIS Living Atlas, Esri Demographics*



Other open data sources

- https://livingatlas.arcgis.com/
- https://hub.arcgis.com/

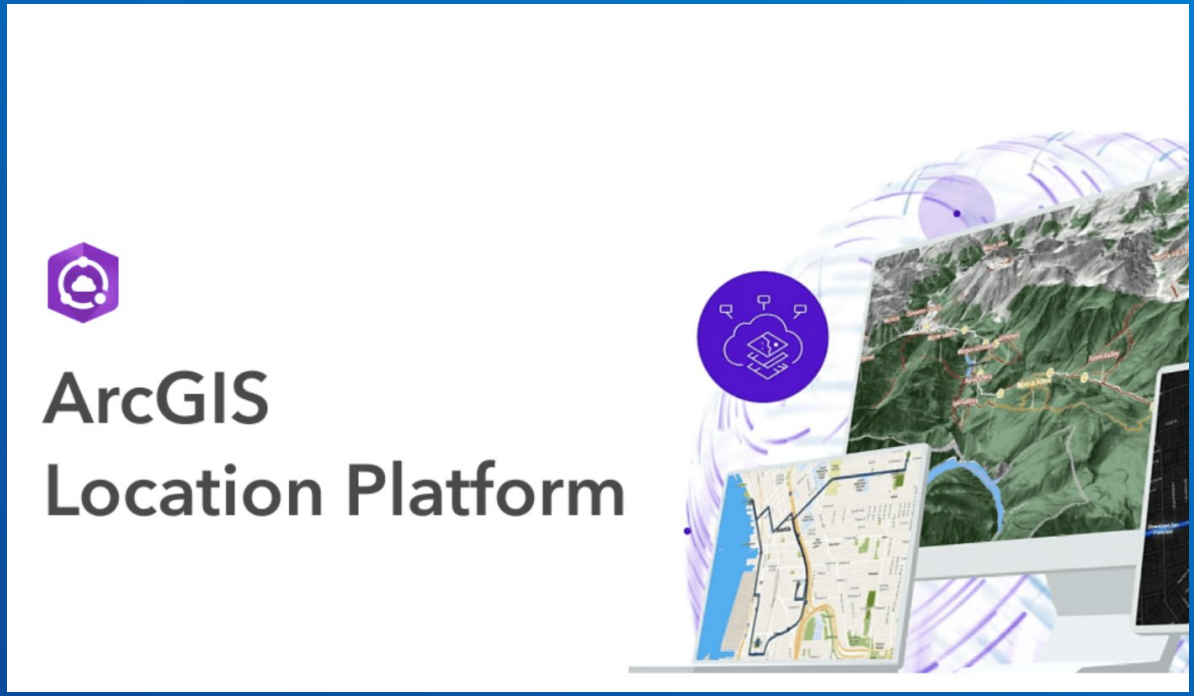
- All free
- Worth checking your area



Accessing ArcGIS open data

Use ArcGIS Location Platform

- Free and open source
- Make a free account: ArcGIS Location Platform
- Access Hub data, Living Atlas data
- Also get data hosting, platform services

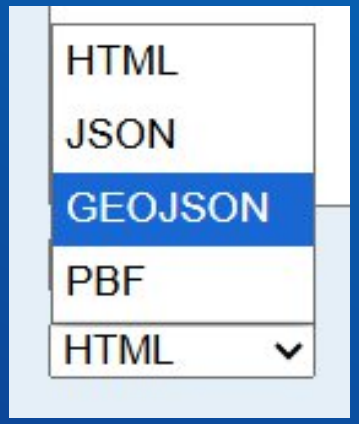


Features: Open data formats

- A world of wonderful standards!
- JSON-FG, GeoParquet, Vector tiles, GeoPackage, Topo-JSON, PMTiles, KMZ.....

- Reality: **GeoJSON everywhere**

- Esri feature services return GeoJSON.
  - Among other, worse options



GeoJSON

Browser-based data rendering

MapLibre GL JS

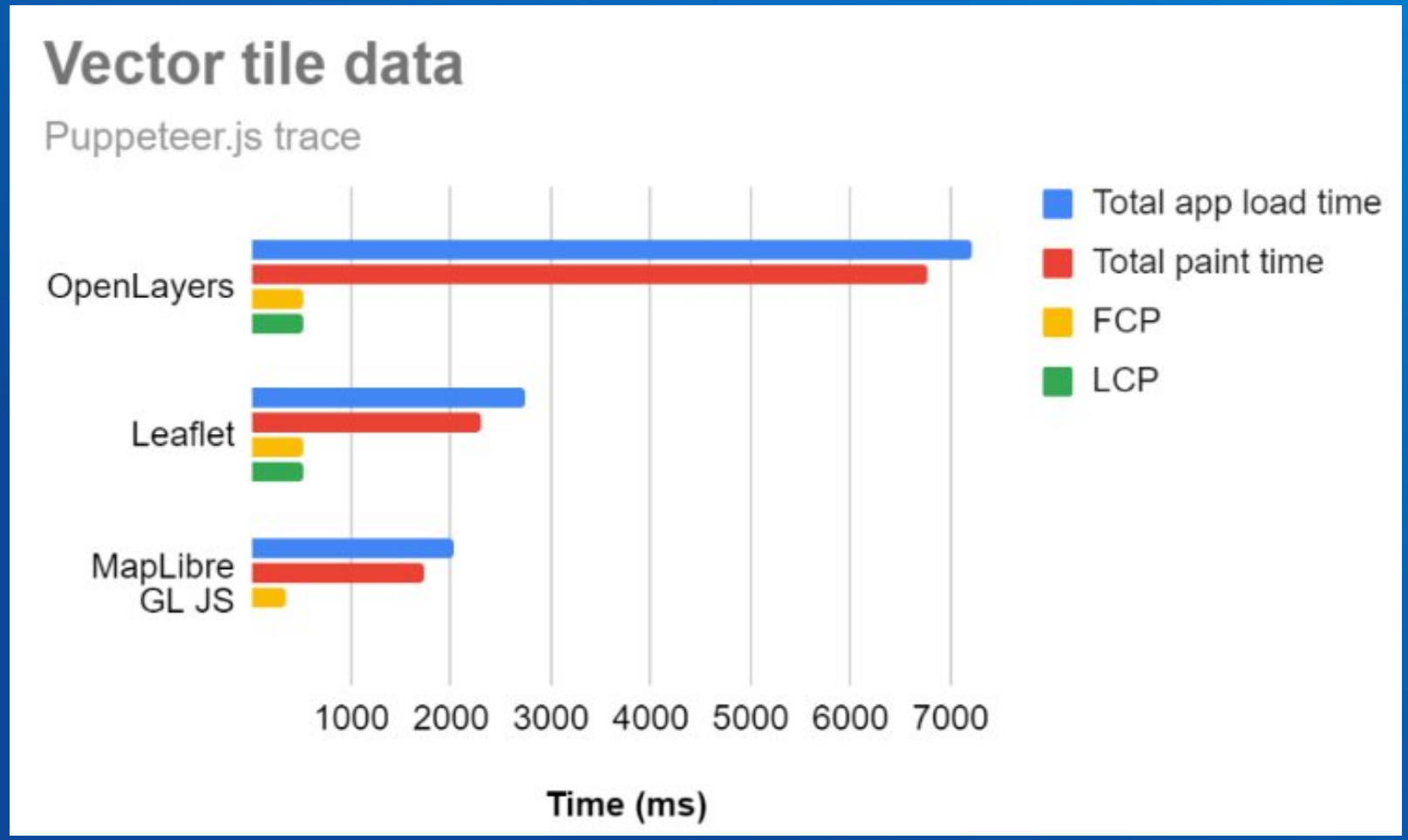
https://maplibre.org/

- Open-source fork of Mapbox v1, 2020
- MapLibre ecosystem: Native SDK, tile server, and more
- Wonderful alternative to Mapbox

- Strong industry support
- Active community
- Growing in popularity



MapLibre GL JS is performant



- Source: my 2023 FOSS4G talk

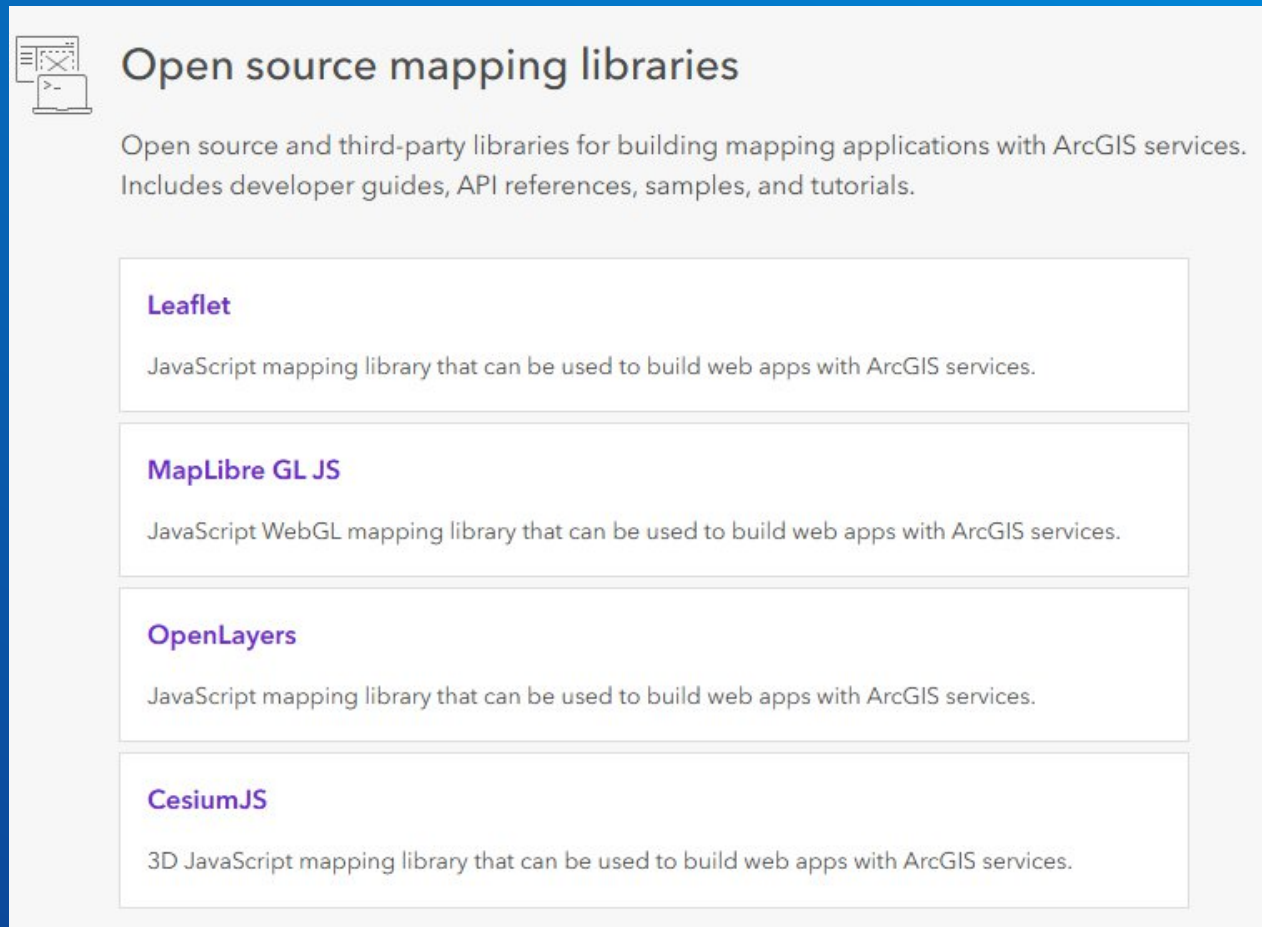
MapLibre

Esri’s open-source ecosystem

- Javascript/Typescript npm packages

- **ArcGIS REST JS**
- **MapLibre ArcGIS plugin**
- Leaflet ArcGIS plugin (Esri Leaflet)
- CesiumJS, OpenLayers support

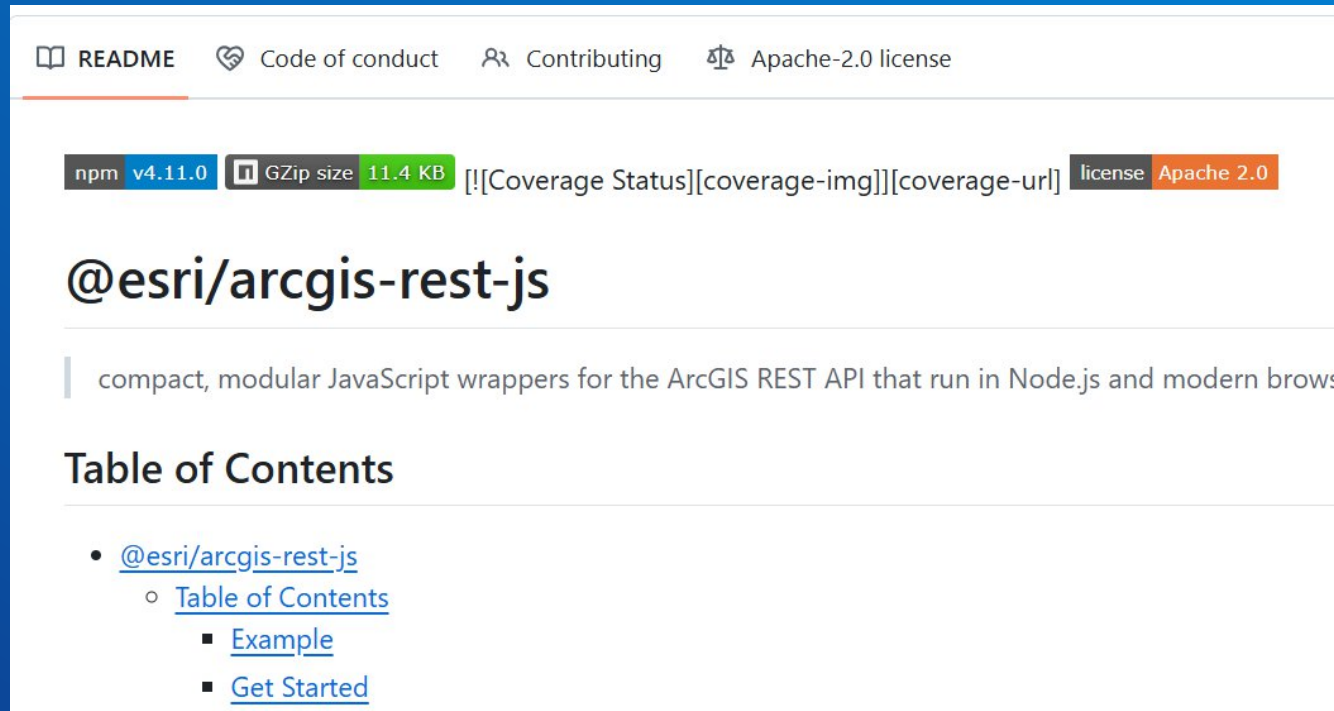
- More plugins coming soon!



ArcGIS REST JS

@esri/arcgis-rest-js

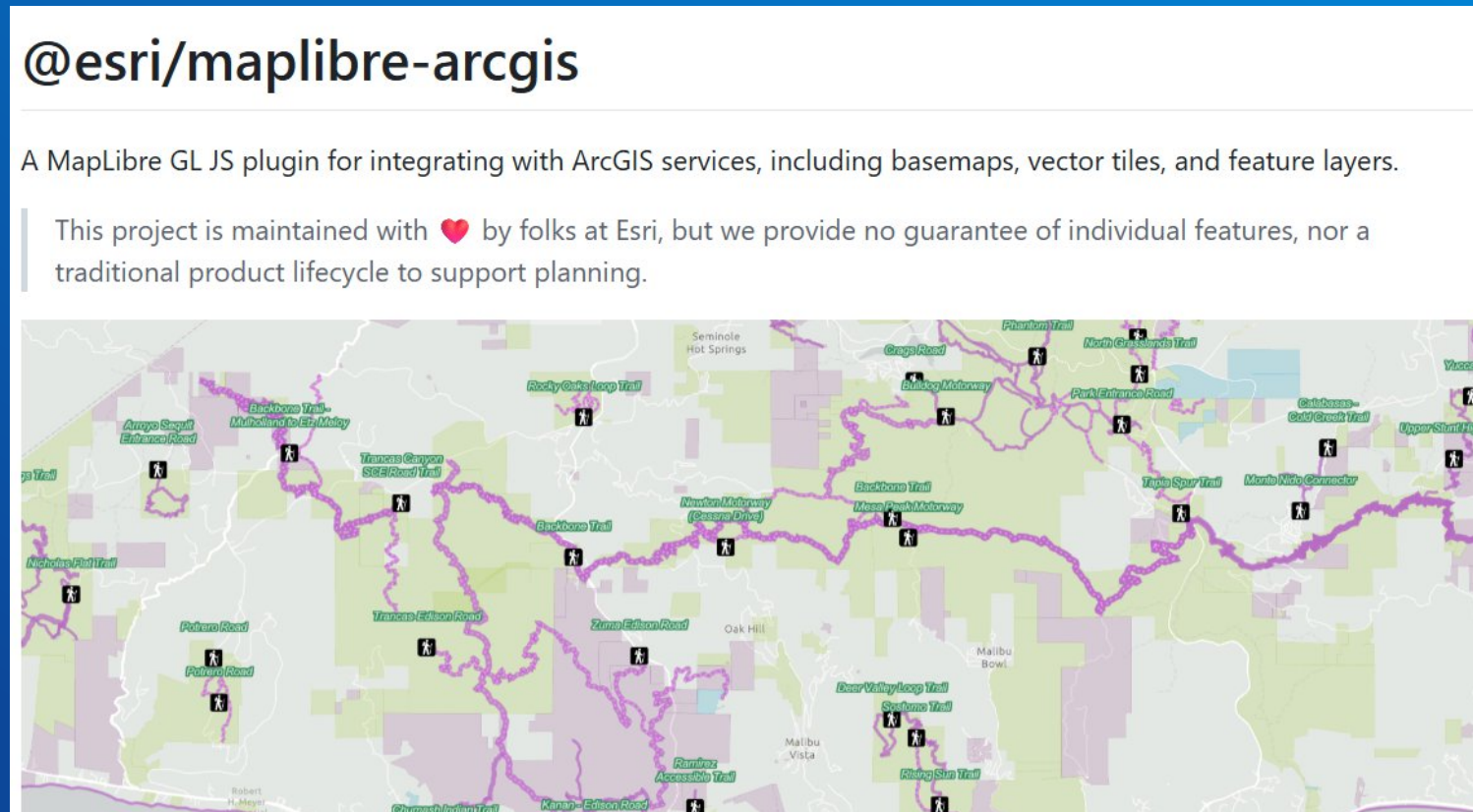
- Typescript monorepo containing 10 packages
- REST API helpers – data services, portal authentication, etc
- Core re-usable logic for other packages
- Also powers ArcGIS Hub!
- LINK



MapLibre ArcGIS plugin

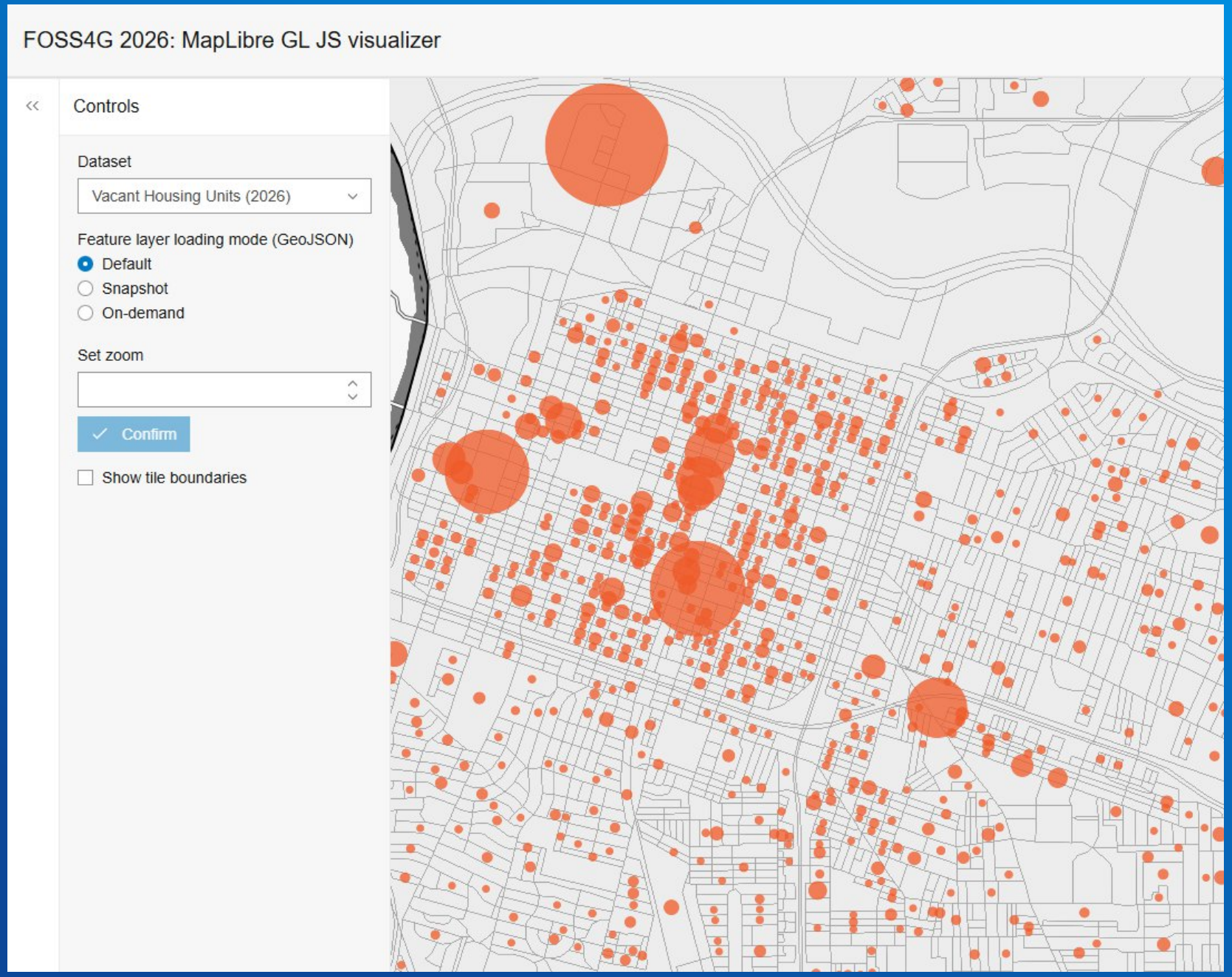
@esri/maplibre-arcgis

- Data service integration with MapLibre GL JS
- Basemap services
- Feature services
- Vector tile services
- Attribution, authentication, etc
- LINK



Demo

- LINK

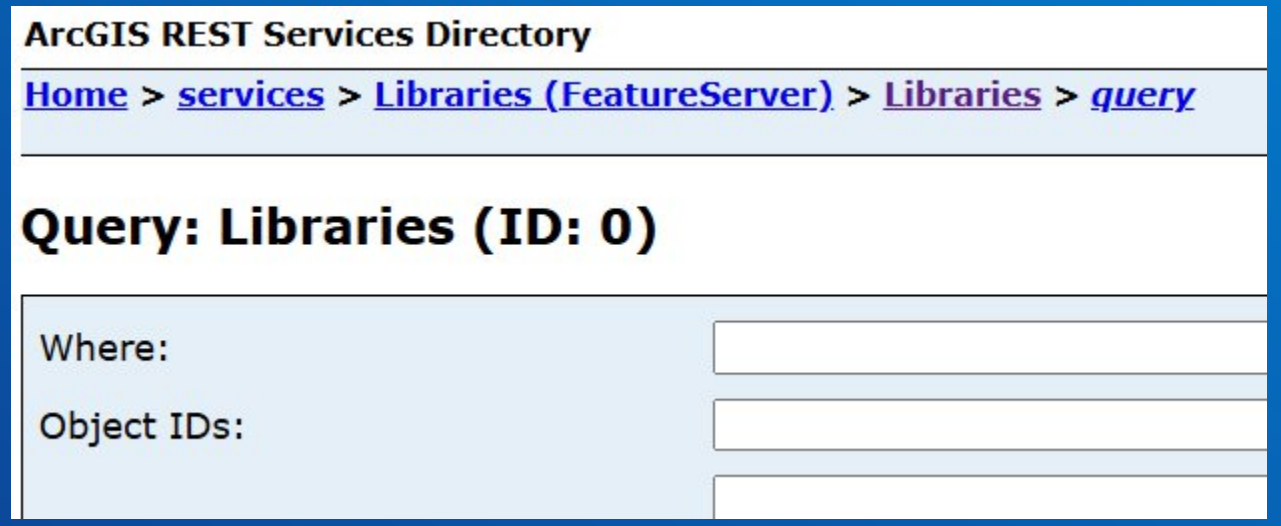

FOSS4G 2026: MapLibre GL JS visualizer
Controls
Dataset
Vacant Housing Units (2026)
Feature layer loading mode (GeoJSON)
Default
Snapshot
On-demand
Set zoom
Confirm
Show tile boundaries

Optimizing Feature service queries

Problem: Esri's Feature Service

MapLibre ArcGIS

- Esri feature services only serve GeoJSON
  - *Spatially Optimized Parquet service also available, but distinct from feature services
- GeoJSON is unoptimized
  - Large file sizes
  - Slow loading times
- 2,000 feature limit per request
- **GeoJSON is still hugely popular**



Solution: FeatureLayer class

MapLibre ArcGIS

- Load feature services in MapLibre
- Lessons learned from ArcGIS Maps SDK for JS
- Huge amount of optimizations
  - Multiple loading strategies
  - Advanced caching strategy
  - Network transfer optimizations
  - Converts EsriJSON to GeoJSON
- “The secret sauce”

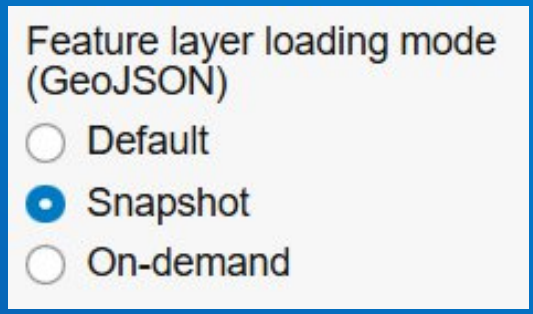




Loading strategy: Snapshot mode

MapLibre ArcGIS

- **Load everything at once**
- Triggers on initial render
- Paginate requests up to 8,000 features
  - Bypasses “2,000 feature limit”
- Easy to implement
- Real advantages for small datasets

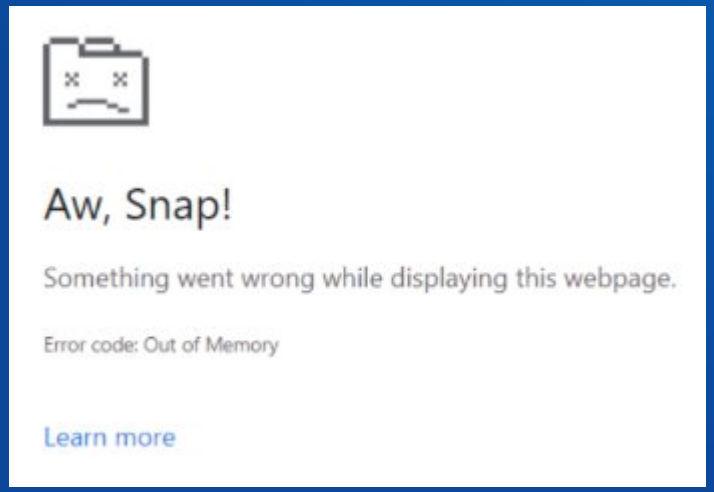


Snapshot mode pros and cons

MapLibre ArcGIS

- + Avoids unnecessary requests
- + Real advantages for small datasets
- + Ideal situation: Get it out of the way early

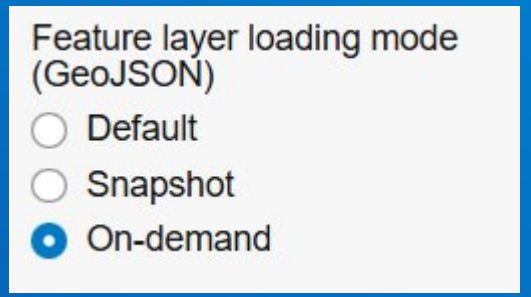
- - Blocks first contentful paint (FCP)
- - Parcels:



On-demand loading strategy

MapLibre ArcGIS

- **Load tiles as you pan and zoom**
- Cache the current view and a bit more
- Optimize geometries at different zoom levels
  - Quantization
- Enables truly large datasets



Quantization parameters

MapLibre ArcGIS

- Handled by FeatureServer
- Simplify the geometry of features
  - Simplified version has fewer vertices, more compact

- Tolerance parameter: Request different versions at different zoom levels
- API reference

Quantization Parameters:


```
{"extent":{"spatialReference":
{"latestWkid":4326,"wkid":4326},"xmin":-121.640625,"ymin":38
.27268853598097,"xmax":-121.2890625,"ymax":38.54816542304657
},"mode":"view","tolerance":0.00017578125}
```

More on-demand optimizations

MapLibre ArcGIS

- Client caching strategies
  - Lagging window method
  - Current zoom and extent
  - Extent-based expiration
  - Lagging window method
- AbortController to deduplicate requests
- FeatureService: Server-side cache

- Hardcoded overflow prevention: 350,000 features on-screen at once
- Dependent on hardware specs

Default FeatureLayer behavior

MapLibre ArcGIS

1. Get service information

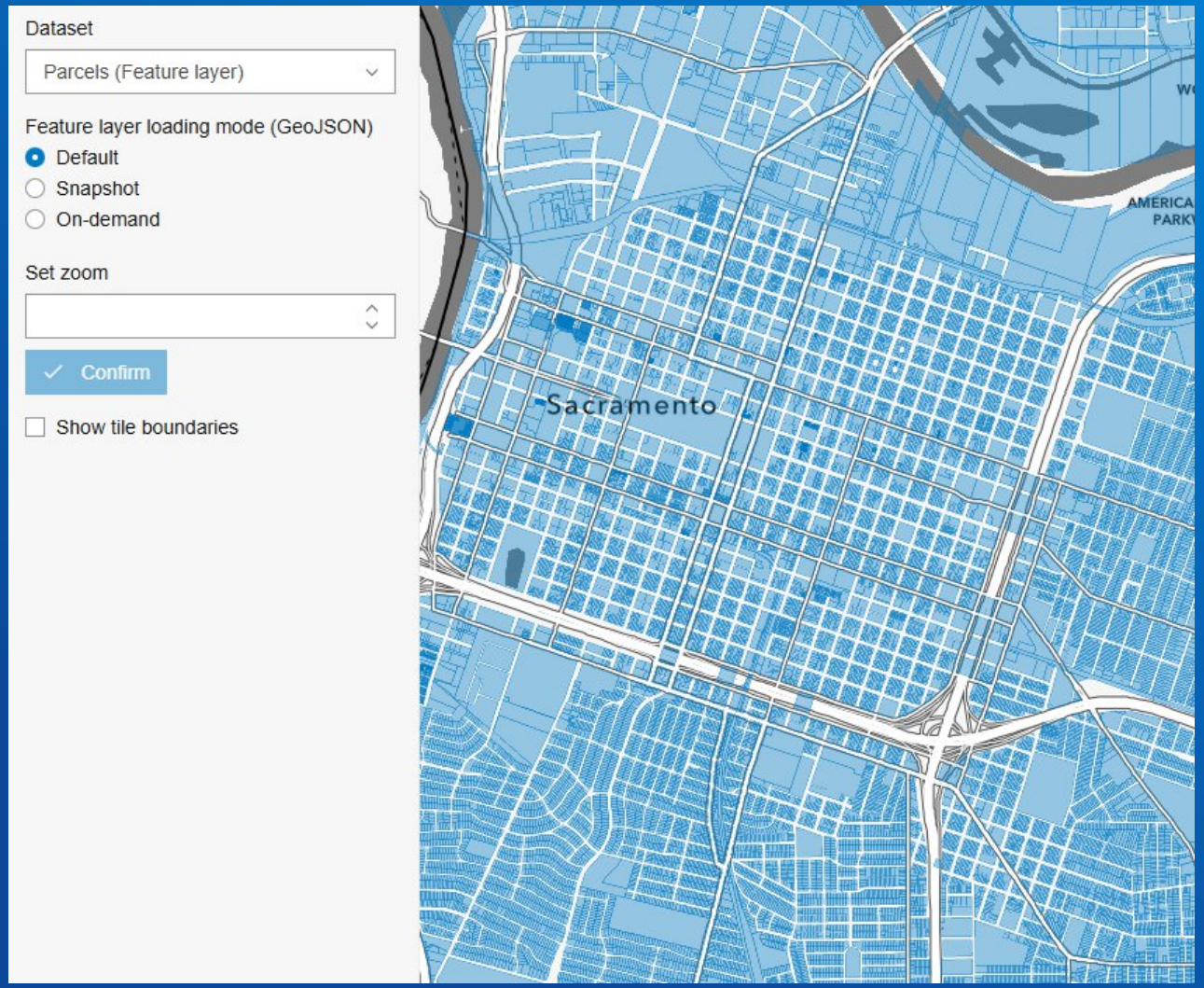
2. Attempt snapshot mode
    1. Check if exceeds 8,000 feature limit (higher for points)
    2. Load everything at once

3. Fall back to on-demand loading

- “loadingMode” parameter can override default behavior

Demo: Snapshot vs On-demand

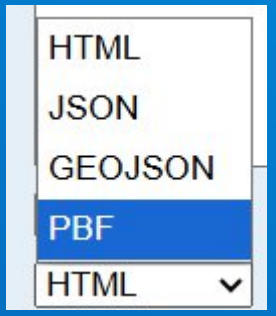
MapLibre ArcGIS



Another optimization: PBF network requests

ArcGIS REST JS

- Request all GeoJSON as highly compressed .pbf files
  - PBF contains EsriJSON, not MVT
- Decode to GeoJSON on client
- ~10x less bandwidth than GeoJSON



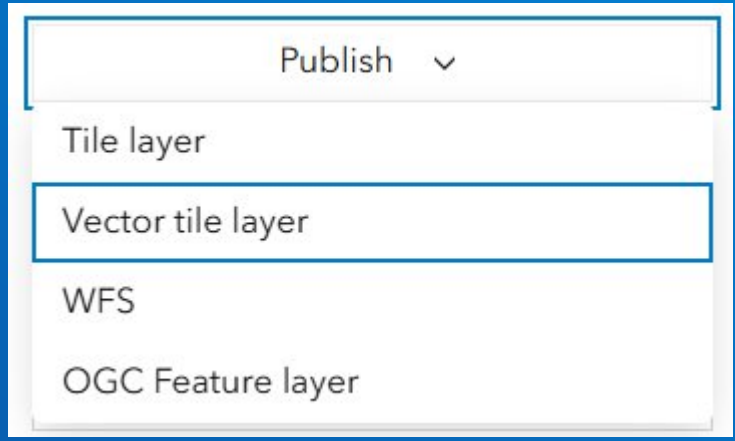
- From feature services: PBF containing EsriJSON :\
- We wrote our own parser: @esri/arcgis-rest-feature-service

- MapLibre ArcGIS workflow:
  - Query EsriJSON PBF -> convert EsriJSON to GeoJSON -> render GeoJSON

The best optimization?

MapLibre ArcGIS

- Stop using GeoJSON...
- Publish vector tiles instead.
  - Identical attributes and geometry
- For existing feature services: Single click



- Or, use a Parquet Feature Layer

The road ahead

Esri’s open source ecosystem

- Implementing this logic in more environments
  - Leaflet
  - CesiumJS
- What other libraries are you interested in?
- What do you want to see in MapLibre ArcGIS?
  - Parquet feature layer support



- Slides and app: https://github.com/gowin20/foss4g-2026-maplibregl-data-visualization



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